

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

20BT503 / 17BT503 – BIOINFORMATICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Outline the different applications of bioinformatics.	10	L*2	1	1,2
	b)	What is needed for chain-termination sequencing?	10	L1	1	1,2
2.	a)	Illustrate on the 5 benefits of the human genome project.	10	L2	1	2,4
	b)	List and elaborate on the methods used to measure protein expression.	10	L1	1	2,4
		Unit – II				
3.	a)	How does the concept of an object-oriented model differ from the concept of an entity in the ER model?	10	L4	2	1,2
	b)	Summarize the difference between primary and secondary biological database.	10	L3	2	1,2
4.	a)	What information can you derive from a PDB file?	10	L2	2	1,2
	b)	What is the difference between Entrez and SRS?	10	L5	2	2,4
		Unit – III				
5.	a)	What is the difference between local alignment and global alignment?	10	L1	3	1,2
	b)	Elaborate on the application of PAM and BLOSUM scoring matrices.	10	L3	3	3,4
6.	a)	What are the advantages of Hidden Markov model?	10	L4	3	4,5
	b)	Given below are two randomly generated sequences: ATCG and TCG; Create a score matrix and deduce the alignment of these two sequences using Needleman–Wunsch algorithm. Assume: Match score +1, Mismatch score -1	10	L2	3	3,5
		Unit – IV				
7.	a)	What is molecular evolution in bioinformatics?	10	L3	4	2,4
	b)	What is the difference between rooted and unrooted phylogenetic tree?	10	L2	4	3,4
8.	a)	With a suitable example, discuss the application of distance based method in tree construction?	10	L2	4	3,5
	b)	Construct an example for a good bootstrap value for phylogenetics.	10	L2	4	3,5
		Unit – V				
9.	a)	Discuss the applications of Seq Lab in Bioinformatics.	10	L2	5	3,5
	b)	How does bioinformatics help in drug designing?	10	L2	5	2,4
10.	a)	Elucidate the steps involved in homology modeling for the protein tertiary structure prediction.	10	L2	5	2,4
	b)	Elucidate the application of systems biology in biotechnology.	10	L2	5	3,4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
